

Delusterants for man-made fibers

Man-made fibers such as rayon, acetate, nylon, and polyester often have a high degree of luster because their relative transparency or their shape reflects considerable light. A few long light rays reflected from a surface give more luster than many short rays, because the short rays are diffused as they cross and blend. Luster may be controlled by selection of fibers or yarn construction techniques. However, when this is not feasible, supplementary substances can be added to the fiber during manufacture. A common system is to introduce pigments into the spinning solution. Titanium dioxide, a white pigment, reduces luster by the breakup of light reflection, thus creating an opaque and dull fiber. The amount of delusturing can be controlled by the amount of titanium dioxide used.

External delusterants may be applied to the surface of fibers or yarns. Barium salts, such as barium sulphate, are the most common. They are applied in a two-bath procedure that results in the formation of an insoluble deposit on and in the fiber. Other external delusterants are applied in a single bath and deposited on the fiber surface. These include china clay, aluminium oxide, zinc oxide, and methylene urea.